Minutia Deviation Tool (MDT)

So the deal is, we're working with the Defense Biometrics and Forensics folks and NIJ on this broader contactless fingerprint research effort, and right now nobody has a good way to actually compare two fingerprint captures side by side and see how much the minutiae points drift between them. Like, you take a contact scan and a contactless scan of the same finger, and you want to know objectively how far off the matching ridge features are from one another — there's just no tool for that today. Researchers are basically eyeballing it or doing something manual and clunky.

What I'm picturing is a pretty focused little desktop utility, nothing fancy, that lets a researcher open up a pair of fingerprint files, see the image with all the minutiae points overlaid on it, and then click on a point on one print and the matching point on the other print, and the tool just calculates the spatial deviation between them and shows it right there. They should be able to click into an individual minutia and see its details too — x, y position, angle, classification, whatever annotations are attached to it.

It'd be nice if on top of that they could filter through the deviations somehow, like mark pairs as they go, keep a running list during the session, and maybe filter the display based on some deviation threshold they type in. And being able to export just the minutiae that pass that filter into a new file would be great too — but honestly that filtering piece is more of a nice-to-have, if it doesn't make it into the first version that's fine, it's not the core ask.

The core ask really is just the open-compare-calculate flow. Whatever they do during a session, they need to be able to save it — both as a kind of session file they can reopen later and pick up where they left off, and as a simple log file, something like a table they could open in Excel, listing the pairs and the deviations calculated. Everything stays local, opened and saved from the user's own machine — there's no need for this to talk to any network or server, that's explicitly not something we want, keep it offline.

Who's going to use this is basically biometrics researchers — academia, government labs, industry partners — people who already know what they're doing with fingerprint data, so we don't need to dumb down the interface or build training wheels into it. This isn't a public-facing product, it's a research instrument that we'll hand out to people who already understand minutiae and EBTS-type files. Because of that, things like user accounts, per-user permissions, fancy security layers — none of that matters here, skip it.

One more thing worth flagging: this is very much a prototype/beta-stage tool, not something we're polishing for general release. It just needs to be stable enough that a knowledgeable researcher can rely on it without us providing ongoing support afterwards — once it's out the door, it's pretty much on its own. So let's not over-engineer it, just get the comparison and measurement piece rock solid first.